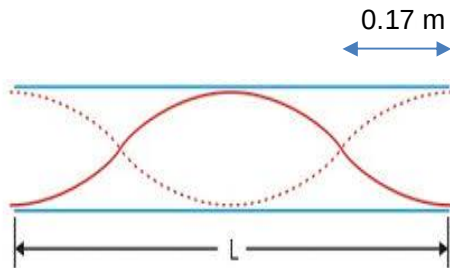
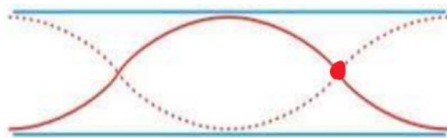


18. Answer: C

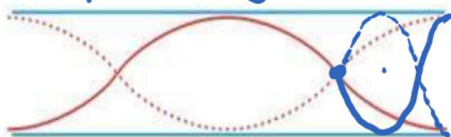


The first node is detected at a distance of 0.17 m from the end. The length of the open tube is 0.68 m. Hence this corresponds to the second harmonic for open tube. Distance of 0.17 m is $\frac{1}{4}\lambda$ from the end.



$$0.17\text{ m} = \frac{1}{4} \text{ of a wavelength}$$

microphone stays at same position. For the new λ' , microphone's position is still a node:



$$0.17\text{ m} = \frac{3}{4} \lambda'$$

$$\lambda' = \frac{4(0.17)}{3}$$

$$\text{hence } f' = \frac{c}{\lambda'} = 1500 \text{ Hz}$$

19. Answer: C